

Juan Manuel Young Hoyos

SOFTWARE ENGINEER, FULL-STACK — TYPESCRIPT · SYSTEMS, REAL-TIME & WEBASSEMBLY

Medellín, Colombia · Open to relocating to Amsterdam (on-site) · English B2+

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Software Engineering — Hiring Team

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MONUMENTAL

Application for Software Engineer, Full-Stack

Dear Monumental Team,

I am applying for the **Full-Stack Software Engineer** role at Monumental. Software that touches the physical world is the reason I am writing: I would rather debug why a robot behaves differently than its simulation than ship another SaaS screen. Building in Amsterdam, on-site, around a robot testing floor, is part of the appeal rather than a cost.

Your stack is **TypeScript and Rust, much of it compiled to WebAssembly** so the UI shares control code with the robots. The closest thing I have done is refactoring and retrofitting a legacy **C++ WebAssembly** application, improving its code quality by **45%** — so I have worked inside a WASM codebase and understand what that shared-stack boundary costs and buys. I should say plainly that I have not written production **Rust**; I have shipped C++, Go, Python and TypeScript, I pick up languages quickly, and your posting says Rust is valuable but not required, so I will be straight with you rather than imply otherwise.

The problems you describe are ones I have versions of. **Telemetry so you can see what every unit is doing:** as Technical Lead at Okorum I stood up logs, traces and metrics across a distributed system and engineered **fault tolerance and resumability** into its queues, so in-flight work survives an unreliable external dependency that goes down without warning — the software equivalent of a flaky link on a site. **Choosing the right abstractions:** I picked the core architecture for that platform and shipped it ahead of a hard government deadline. **Real-time and protocols:** I drove an **MQTT** architecture at ArcDesign with device **simulations in Python** before integrating the logic for near real-time behaviour, and I have built event-driven systems on Kafka and RabbitMQ.

On **LLM agents:** I run several in parallel — **Claude Code** for building, a separate pass for code review, and others for investigation and documentation — and I am usually the person colleagues ask when an agent does something inexplicable. I treat their output as a draft to be validated, not a result.

On foundations: I hold a B.S. in Computer Science and Engineering, I built a geohash-and-hashmap spatial algorithm for scalable geolocation, and I have done 3D engine work in Godot and Unity, so transforms and linear algebra are familiar ground. I am comfortable on Linux, SSH-ing into a machine to work out why something is not behaving.

Finally, ownership: I have repeatedly been the person who drives a project start-to-finish without someone managing it, in chaotic conditions and against deadlines that did not move. Thank you for your consideration.

Sincerely,

Juan Manuel Young Hoyos

Attached: Resume — Juan Manuel Young Hoyos.pdf